

Darren Lee

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EDUCATION

University of California, Berkeley

May 2028

B.S. Electrical Engineering and Computer Science, Minor in Materials Science and Engineering

Berkeley, CA

- **Relevant Coursework:** SICP, Linear Algebra + Diff Eq, Data Structures, Signals and Systems, Physics for Scientists and Engineers

Ruben S. Ayala High School

Chino Hills, CA May 2025

- GPA: 4.61, ACT: 35, 16 AP Classes (All 5's and 4's), Class Rank: 7/672
- **Relevant Coursework:** AP Calc AB/BC, Stats, Physics 1 and C Mech/E&M, CSA, CSP

SKILLS AND AWARDS

Technical: Circuit design, Circuit analysis, PCB design (KiCad), Spice simulation, Bench instrumentation, Soldering, Prototyping, 3D printing, C++, Python, Arduino, Java, SQL, Scheme, Woodshop

Awards: Regents' and Chancellor's Scholar, CVUSD Superintendent's Award, Gold President's Volunteer Service Award

EXPERIENCE

Engineering Solutions at Berkeley

Sept 2025 – Present

Electrical and Software Engineering Consultant

Berkeley, CA

- Selected into UC Berkeley's first student-run engineering consulting organization (competitive, all-majors application) to take on a client-sponsored hardware project, delivering a fully documented, client-ready solution
- In charge of the electrical and embedded-software scope on a small cross-disciplinary team, collaborating with other engineers to progress through semester-long projects following ESB's full consulting workflow

Fusion Dynamics

June 2026 – Aug 2026

Electrical Engineering Intern

Oakland, CA

- Scoped a drone-agnostic autonomous docking station that concentrates landing intelligence in the dock rather than the aircraft, evaluating fiducial-marker landing, IR beacon tracking, and Betaflight command uplink
- Designed a four-LED 850 nm IR beacon board in KiCad, computationally validating candidate emitter geometries for rotational distinctness so the pose solver could not lock onto a wrong orientation
- Assisted in designing a 4-layer 95 x 55 mm structural frame PCB serving as both the top plate and the power distribution board of a sub-250 g drone, consolidating mechanical mounting and current paths into a single part

UC Davis Undergraduate Level Research Program

July 2024 – Aug 2024

Researcher – Physics & Engineering Cluster

Davis, CA

- Applied nuclear and electrical engineering principles to analyze real-world systems, performing quantitative evaluations of fission/fusion processes, critical mass requirements, reactor constraints, and electro-optical behavior
- Conducted analytical work modeling foundations of physics, deriving and validating equations from Newtonian mechanics, quantum mechanics, and special relativity to evaluate alignment between models and observations
- Executed laboratory investigations using interferometry, laser systems, optical waveguides, and fiber-optic setups to characterize electromagnetic propagation, assess signal transmission behavior, and interpret experimental results

PROJECTS

RF Generator for Quadrupole Ion Guide | *Evan R. Williams Research Group (via ESB)*

Feb 2026 – June 2026

- Designing a custom high-voltage RF generator to drive a quadrupole ion guide in a mass spectrometer, targeting stable RF waveforms for precise ion confinement
- Researched and benchmarked commercial step-up transformers across multiple manufacturers (Coilcraft, Lundahl, Hammond), evaluating bandwidth, voltage gain, and power handling for high-voltage RF drive
- Custom-wound and bench-tested multiple transformer prototypes on Type 43 NiZn ferrite toroids and Material 77 U-cores, comparing turns ratios and core configurations to optimize gain and resonant behavior

Precision Focus Actuator | *Malin Space Science Systems (via ESB)*

Sept 2025 – Dec 2025

- Designed and automated a focus actuator for space camera applications used by NASA rovers and other commercial aerospace customers, requiring +/- 2 mil actuation with high precision
- Developed the control system (DRV8834 microstepped motor, dual limit switches, ratiometric hall sensor) and implemented homing, midpoint detection, and adaptive backlash algorithms in C++ for micromechanical actuation
- Validated electrical functionality through 3D-printed iterations, performing step-displacement calibration, hall sensor median filtering tests, limit switch reliability checks, and closed-loop backlash compensation verification